# Project Overview

# SmartWilds Data Analysis

## Research questions:

1. What are the characteristic workloads of multimodal sensors? What are their SLOs? What hardware is required to support them?
    a. Quantify data arrival rates - both ‘useful’ and empty images/files
    b. What type of hardware would be needed to just report the *information,* not just collect the data, i.e. report # of species, # of animal sightings
2. When does it make sense to deploy AI at the edge? How to measure this?
    a. When can we trust models to make decisions about data storage? How much longer can sensors be deployed with on-edge data management? Tradeoffs between power/storage for data vs power/storage for edge AI models. How much man hours could be saved with edge computing
    b. What do we mean by data management at the edge?
3. How well do different modalities capture wildlife biodiversity?

## Project plan

1. **Annotate data (Nov - March)**
    a. Start with generating heatmaps from June data (Jupyter notebook) to show Dan the species land use by modality (Nov - Dec)
        i. Bharath - camera traps
        ii. Tanishka - acoustics
        iii. Anirudh - drones
    b. Annotate the remainder of the datasets
        i. Use megadetector + bioclip for cam traps
        ii. Bioacoustics
        iii. Citizen science
        iv. Drone
        v. GPS
2. **Analyze traces for systems paper (Feb-Mar)**
    a. Model multi-modal workloads (using techniques from SEC paper)
        i. SLOs
        ii. Inference speed
        iii. % of requests met
        iv. Communication overhead
    b. Characterizing hardware requirements for next gen of distributed, multi-modal remote sensing
    c. Quantify storage costs of collecting ALL the data vs analysis in the field
        i. Storage costs
        ii. Man hours

    d. **SEC deadline: May**
3. **Use traces to quantify biodiversity metrics (Apr - Sept)**
    a. How well do different sensor modalities capture wildlife biodiversity?
    b. Species richness, species density, landuse
    **c. Write paper with ABC team on this topic**
    d. Prep dataset for release

**Team**

- Leads: Jenna & Anirudh
- Grad students: Rugved (systems experiments), Tanishka (bioacoustic)
- Advisors: Chris, Tanya, Hari, Harry
- Elizabeth + Net (data and software)
- Imageomics undergrads
- Ecology collaborators (ABC)
    - Timm Hauke - camera traps species analysis

PAPER 1: SYSTEMS (SEC)

- Title: "Characterizing and Optimizing Multimodal Wildlife Monitoring Workloads: From Edge Computing to Biodiversity Insights"
- Lead: Jenna
- Focus: Workload characterization, edge computing trade-offs, system design
- Audience: Computer systems researchers, edge computing community
- Key Contribution: Quantify real-world ecological sensor workloads and when edge AI deployment is cost-effective

PAPER 2: CV BENCHMARK DATASET (KDD)

- Model benchmark for animal detection
- Lead: Anirudh
- Datasets + OWL

PAPER 3: ECOLOGY (Methods in evolution and ecology)

- Title: "Multi-Sensor Fusion for Biodiversity Assessment: Accuracy, Complementarity, and Cost-Effectiveness Trade-offs"
- Lead: Jenna
- Focus: Sensor effectiveness, multi-sensor integration, ecological recommendations
- Audience: Ecologists, conservation practitioners, monitoring program designers
- Key Contribution: Validate sensor accuracy with ground truth, demonstrate multi-sensor fusion improves biodiversity estimates
- Publish dataset?

# Ecology Paper

# How well do different sensor modalities capture wildlife biodiversity?

Coauthors: ABC (Justin, Laura, Sara, Kaitlyn, **Oisin**, David, Marta), Timm

| **Sensor** | **Biodiversity Metric** | | |
|---|---|---|---|
| | **Species Richness** | **Species Abundance** | **Land use** |
| Camera trap | | | |
| ARU | | | |
| Drone | | | |
| GPS | | | |
| Citizen science | | | |

## Species Richness

- **What it measures:** The number of different **species** present in an area
- **Unit:** Count of species (no unit)
- **Example:** A forest plot contains 12 bird species
- **Key point:** Doesn't tell you anything about how many individuals of each species exist

## Abundance

- **What it measures:** The number of **individuals** of a species (or total individuals)
- **Unit:** Count of individuals (no standardization by area)
- **Example:** There are 450 robins in the forest
- **Key point:** Absolute number without reference to space

## Density

- **What it measures:** The number of individuals per unit **area** or volume
- **Unit:** Individuals per area (e.g., birds/hectare, fish/$m^3$)
- **Example:** There are 15 robins per hectare
- **Key point:** Abundance standardized by area, allowing spatial comparisons

Use **JSDM** - Laura P and Oisin to estimate habitat usage for species by modality
What happens when you combine all 3 modalities?

First steps: species ID and counting

References:

- Assessing the camera trap methodologies used to estimate density of unmarked populations
- Understanding co-occurrence by modelling species simultaneously with a Joint Species Distribution Model (JSDM)
-

# Systems Paper

**Overleaf:** https://www.overleaf.com/3156758585vrqbvkxtnccm#202e2d

**Title idea**

- Optimizing Edge-Cloud Trade-offs for Sparse-Event Detection: Workload Characterization and Performance Modeling
- When to Deploy Edge AI: Performance Analysis and Decision Framework for Sparse-Event Detection Systems

Use methodology established in SEC to quantify multi-modal workloads in ecology; Quantify tradeoffs for edge computing vs cloud

Pitch: We present the first comprehensive characterization of sparse-event detection workloads and provide an evidence-based framework for edge deployment decisions. Using environmental monitoring as a representative case study, we quantify arrival patterns, performance trade-offs, and optimal edge-cloud configurations. Our open workload traces and decision tools enable the systems community to benchmark edge computing designs and optimize resource allocation for this emerging class of applications/

**Outline**

- **[Context - HPC/Systems problem]**: Emerging edge sensing applications generate massive data volumes, yet most data contains no events of interest. Understanding these sparse-event workloads is critical for designing efficient edge-cloud systems.
- **[Gap - Why this matters to SC]** However, no prior work has quantitatively characterized the arrival patterns, temporal dynamics, and processing requirements of real-world multimodal sparse-event detection workloads at scale.
- **[Approach - Systems contribution]** We present the first comprehensive workload characterization of multimodal edge sensing systems, using a 30-day environmental monitoring deployment as a representative sparse-event detection case study.
    - We quantify data arrival distributions, burstiness metrics, temporal patterns, and multi-modal heterogeneity across camera, acoustic, aerial, and GPS sensors.
- **[Key Results - Performance metrics]** Our analysis reveals:
    - (1) extreme burstiness (CV=XX) with XX% empty data creates storage bottlenecks,
    - (2) edge AI filtering reduces data movement by XX× with <XX% accuracy loss,
    - (3) edge inference power (XXmW) is XX lower than transmission (XXW), making edge deployment cost-effective after XX days,
    - (4) multi-modal fusion reduces processing latency from days to minutes while improving detection accuracy XX%.
- **Contribution - Tools for community]** We provide:
    - (1) open workload traces enabling edge computing benchmarking,
    - (2) EdgeBio decision framework for edge vs cloud deployment,
    - (3) performance models for power-accuracy-latency trade-offs,
    - (4) simulation tools for evaluating edge strategies before deployment.

- [Broader impact] Our methodology generalizes to traffic monitoring, industrial inspection, healthcare sensing, and other sparse-event detection domains.

# draft data analysis

PROJECT SUMMARY: MULTI-SENSOR BIODIVERSITY MONITORING COST-EFFECTIVENESS ANALYSIS

## PROJECT OVERVIEW

Objective: Compare the cost-effectiveness of different sensor modalities (camera traps, acoustic recorders, drones, GPS collars, citizen science) for measuring biodiversity metrics (species richness, abundance, land use) in a managed park with known ungulate populations.

Key Innovation: Systematically evaluate trade-offs between data quality, energy consumption, and storage requirements across different sampling strategies.

Ground Truth: Known populations of 3 ungulate species provide validation benchmarks for abundance estimates.

## RESEARCH QUESTIONS

1. Which sensor/setting combinations most accurately estimate ungulate abundance?
2. How do different ARU recording schedules affect bird species richness detection?
3. What are the energy and storage costs for each approach?
4. What is the optimal sampling strategy for different monitoring objectives and budget constraints?

## DATA COLLECTION SUMMARY

| Sensor Type | Settings Tested | Primary Target |
|---|---|---|

---|---|---
Camera Traps | • High-frequency (1s delay) | Ungulates (3 species)
| • Medium-frequency (30s delay) | Incidental birds
| • Low-frequency (5min delay) |
| • Video mode |
| • Time-lapse |
---|---|---
ARUs | • Continuous recording | Birds (richness)
| • 10% duty cycle | Ungulate vocalizations
| • Dawn/dusk targeted |
| • Acoustic trigger |
---|---|---
Drones | • Standard aerial survey | Ungulates
| | Some birds (waterfowl)
---|---|---
GPS Collars | • 4-hour fix rate | Tagged ungulates

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ANNOTATION SCHEMA
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**CORE FIELDS (ALL DATA TYPES) - REQUIRED**

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✓ date - ISO format: YYYY-MM-DD
✓ time - 24hr format: HH:MM:SS (local timezone)
✓ location - Lat/long or site ID
✓ species - Common name + scientific name
✓ count - Number of individuals (for each species if multiple)
✓ energy_level - Battery % at time of capture (from EXIF/metadata)
✓ confidence - Annotator certainty: high/medium/low
✓ habitat_type - Forest/meadow/wetland/edge (for land use analysis)
✓ detection_quality - Image/audio quality: good/fair/poor
✓ annotator_id - Who labeled this (for inter-rater reliability)

ADDITIONAL FIELDS BY DATA TYPE

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Camera Traps:
- image_id (unique identifier)
- trigger_mode (motion/time-lapse)

- bbox_coordinates (x, y, width, height for MegaDetector/BioClip)
- individual_id (if animal identifiable by markings/antlers)
- behavior (optional: standing/feeding/running/bedded)
- age_class (optional: adult/juvenile)
- sex (optional: male/female/unknown)

ARUs:
- recording_id (unique identifier)
- detection_start_time (seconds from recording start)
- detection_end_time (seconds from recording start)
- call_type (song/call/alarm)
- background_noise_level (none/low/medium/high)
- overlap_with_other_species (yes/no - affects detection)

Drones:
- flight_id (unique identifier)
- altitude_m (flight altitude)
- image_id_in_sequence (frame number)
- bbox_coordinates (for animal detection)
- movement_visible (yes/no - helps distinguish animals from objects)

GPS Collars:
- individual_id (collar/animal ID)
- fix_quality (GPS accuracy: good/fair/poor)
- activity_sensor (if available: active/resting)

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ANNOTATION TASKS BREAKDOWN
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PHASE 1: DATA PREPARATION (WEEK 1)
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Task 1.1: Organize Raw Data
- Assigned to: Project lead
- Deliverable: Folder structure with all files renamed consistently
- Time: 2-3 days

Suggested folder structure:
project_data/
├── camera_traps/

```
│   ├── high_frequency/
│   ├── medium_frequency/
│   ├── low_frequency/
│   └── video_mode/
├── arus/
│   ├── continuous/
│   ├── duty_cycle/
│   └── dawn_dusk/
├── drones/
│   └── survey_flights/
└── gps/
    └── collar_data/
```

Task 1.2: Extract Metadata
- Assigned to: 1 student with coding experience
- Deliverable: CSV with basic metadata (date, time, location, energy from EXIF)
- Time: 2-3 days
- Notes: Can be partially automated using ExifTool for images

Task 1.3: Create Annotation Templates
- Assigned to: Project lead
- Deliverable: Spreadsheet templates or annotation software setup
- Time: 1 day

PHASE 2: CAMERA TRAP ANNOTATION (WEEKS 2-4)
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Task 2.1: Run MegaDetector (Automated)
- Assigned to: 1 student with Python experience
- Deliverable: Filtered dataset with animal detections only
- Time: 2 days
- Expected outcome: Remove 70-90% of empty images

Task 2.2: Species Identification - Ungulates
- Assigned to: 2-3 students
- Training needed: 2-hour session identifying 3 ungulate species
- Annotation rate: ~100-150 images/hour after training
- Total images to annotate: [X] images
- Estimated time: [X] hours
- Quality control: 10% overlap between annotators to check agreement

Annotation Protocol:
1. Open image in annotation tool

2. For each animal detected:
   - Draw bounding box around animal
   - Select species from dropdown
   - Count number of individuals
   - Note confidence level
   - Record age/sex if visible
3. Note habitat type visible in image
4. Rate image quality
5. Save and move to next

Task 2.3: Incidental Bird Detections
- Assigned to: 1 student with bird ID experience
- Note: Only ground-dwelling birds likely visible
- Time: 1-2 days (low priority)

PHASE 3: ARU ANNOTATION (WEEKS 2-5)
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Task 3.1: Run BirdNET (Automated)
- Assigned to: 1 student
- Deliverable: Initial species detections with timestamps
- Time: 1-2 days (mostly computer time)
- Expected outcome: Automated bird species list

Task 3.2: Validate BirdNET Detections
- Assigned to: 2 students with birdsong experience
- Annotation rate: ~50 detections/hour
- Sample size: 500 stratified detections
- Estimated time: 10-15 hours per student

Validation Strategy:
- 100 high-confidence detections (>0.8): Quick verification
- 200 medium-confidence (0.4-0.8): Careful review
- 100 low-confidence (0.25-0.4): Detailed analysis
- 100 random sample: Unbiased check

Task 3.3: Annotate Ungulate Vocalizations (Custom Model Training)
- Assigned to: 1 graduate student (vocalization specialist)
- Annotation needed:
  * Minimum: 100 positive + 200 negative = 300 clips (15-25 hours)
  * Recommended: 300 positive + 500 negative = 800 clips (40-60 hours)
- Deliverable: Training dataset for acoustic model

Annotation Protocol for Ungulate Calls:
1. Listen to 5-10 second audio clip
2. View spectrogram
3. Classify: Contains ungulate vocalization? (Yes/No)
4. If yes:
   - Mark start/end time of call
   - Note call type if identifiable (alarm/contact/mating)
5. Rate audio quality and background noise
6. Note confidence level

PHASE 4: DRONE ANNOTATION (WEEK 3)
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Task 4.1: Run Custom Detection Model (Automated)
- Assigned to: Student with model
- Deliverable: Bounding boxes around detected animals
- Time: 1 day

Task 4.2: Species Verification with BioClip
- Assigned to: 1-2 students
- Training: 1-hour session on identifying ungulates from aerial view
- Annotation rate: ~200 detections/hour
- Total detections: [X]
- Time: [X] hours

Task 4.3: Manual Review of Low-Confidence Detections
- Assigned to: Same students
- Focus: Verify/correct species IDs with confidence <0.5
- Time: 2-3 hours

PHASE 5: GPS DATA PROCESSING (WEEK 2)
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Task 5.1: Clean GPS Data
- Assigned to: 1 student
- Deliverable: Filtered GPS points (remove outliers, poor fixes)
- Time: 1 day

Task 5.2: Annotate Habitat Types
- Assigned to: 1 student with GIS experience
- Method: Overlay GPS points on habitat map
- Deliverable: Each GPS point tagged with habitat type

- Time: 1-2 days

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QUALITY CONTROL PROCEDURES
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Inter-Annotator Reliability
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- 10% overlap: Same images/recordings annotated by 2+ students
- Target agreement: >90% for species ID, >80% for counts
- Resolution: Project lead reviews disagreements

Expert Validation
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- Random 5% sample: Reviewed by expert ecologist
- 100% of uncertain cases: Flagged low-confidence annotations reviewed

Automated Checks
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- Flag impossible combinations (e.g., aquatic birds in forest habitat)
- Flag count outliers (e.g., >50 individuals in single image)
- Check for missing required fields

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TRAINING MATERIALS NEEDED
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For All Students:
1. Project overview presentation (30 min)
2. Annotation software tutorial (30 min)
3. Data entry best practices (15 min)

Camera Trap Team:
4. Ungulate species ID guide with example photos
5. Bounding box drawing tutorial (15 min)
6. Practice dataset (50 images to annotate with answer key)

ARU Team:

7. Bird song recognition basics (1 hour)
8. Spectrogram reading tutorial (30 min)
9. Reference calls library for common species
10. Ungulate call recognition (for vocalization student)

Drone Team:
11. Aerial ungulate identification (examples from above)
12. Common false positives (rocks, logs, shadows)

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ANNOTATION TOOLS
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Recommended Software:
- Camera traps: Label Studio (free, web-based, handles bounding boxes)
- ARUs: Raven Lite (free, standard for bioacoustics) or Audacity
- Drones: Label Studio or CVAT
- Data management: Google Sheets or Airtable (real-time collaboration)

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TIMELINE SUMMARY
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| Week | Phase | Student Hours Needed |
|---|---|---|
| 1 | Data preparation | 20 hours |
| 2-4 | Camera trap annotation | 60-100 hours |
| 2-5 | ARU annotation | 50-80 hours |
| 3 | Drone annotation | 10-20 hours |
| 2 | GPS processing | 16-24 hours |
| TOTAL | | ~156-244 hours |

Recommended team: 4-6 undergraduate students (10-15 hrs/week each) +
1 graduate student (ungulate calls)

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BIODIVERSITY MEASURES: WILL THIS SCHEMA SUPPORT ANALYSIS?
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WHAT YOU CAN MEASURE:
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| Metric | Supported? | Requirements |
|---|---|---|
| Species Richness | YES | Species field + sample coverage |
| Species Abundance | YES | Count + detection probability modeling |
| Species Density | YES | Count + location + survey effort |
| Land Use/Habitat Selection | YES | Location + habitat_type field |
| Activity Patterns | YES | Date + time fields |
| Detection Probability | YES | Repeated surveys + confidence + quality |
| Cost-Effectiveness | YES | Energy_level + storage metadata |

CRITICAL ADDITIONS NEEDED:
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1. habitat_type - Essential for land use analysis
2. confidence - Important for filtering and uncertainty quantification
3. detection_quality - Affects detection probability
4. annotator_id - Needed for inter-rater reliability

FOR ROBUST STATISTICAL ANALYSIS, ALSO CONSIDER:
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5. survey_effort - How long was camera active? Minutes of audio analyzed?
6. weather_conditions - Affects detectability (optional but valuable)
7. individual_id - If animals individually identifiable (enables
   mark-recapture = more accurate abundance)

================================================================================
DATA VALIDATION WORKFLOW
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Raw Data

↓
Automated Processing (MegaDetector, BirdNET, etc.)
↓
Student Annotation (following protocols above)
↓
10% Inter-Annotator Check
↓
5% Expert Validation
↓
Automated QC Checks (outliers, missing fields)
↓
Final Clean Dataset
↓
Statistical Analysis

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EXPECTED DELIVERABLES
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From Annotation Phase:
1. Annotated camera trap dataset (species, counts, bounding boxes)
2. Validated ARU bird species list (BirdNET + manual verification)
3. Training dataset for ungulate vocalization model (300-800 labeled clips)
4. Annotated drone dataset (species, counts)
5. Processed GPS dataset (habitat-tagged locations)

From Analysis Phase:
6. Comparison table (all sensors × all metrics)
7. Cost-effectiveness analysis (species per GB, species per battery)
8. Validation report (estimates vs. ground truth for 3 ungulate species)
9. Recommendations for optimal sampling strategies
10. Manuscript/report

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RISK MITIGATION
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Risk | Impact | Mitigation

---|---|---
Student turnover mid-project | High | Front-load training, 20% buffer
Low inter-annotator agreement | Medium | More training, clear examples
Automated models underperform | Medium | Budget more manual annotation time
Large data volume overwhelming | High | Prioritize high-value subsets first
Ambiguous species IDs | Low | "Unknown" category + expert review

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NEXT STEPS

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IMMEDIATE (THIS WEEK):
1. Finalize annotation schema (add recommended fields)
2. Set up folder structure and annotation software
3. Extract metadata from all files
4. Create training materials (species guides, tutorials)
5. Recruit students

WEEK 1:
1. Onboard students (training sessions)
2. Run automated pipelines (MegaDetector, BirdNET)
3. Assign annotation tasks
4. Set up weekly check-in meetings

ONGOING:
1. Monitor annotation progress
2. Weekly inter-annotator reliability checks
3. Resolve questions/ambiguous cases
4. Begin preliminary analysis as data becomes available

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QUESTIONS FOR CLARIFICATION

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Before students start, please confirm:

1. Data volume: How many total images/recordings do we have?
2. Habitat map: Do you have existing habitat classifications for the park,

or do students need to annotate this from scratch?
3. Ground truth details: Do you have exact counts for all 3 ungulate species, or ranges? When was last census?
4. Budget: Any constraints on student hours or software licenses?
5. Timeline: Hard deadline for analysis completion?
6. Authorship: Will students be co-authors or acknowledged contributors?

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END OF PROJECT PLAN
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# Data Analysis

# Data Analysis

**Step 1: Generate annotations for each modality - in progress**

A CSV containing the following:

photo/frame/recording - datetime - **location** - SPECIES A Count - SPECIES B Count - …

Coco-style json format for images
Bboxes
Record the annotation method

Bharath - cam traps
Anirudh - drone
Tanishka - bio-acoustoc
Jenna - citizen science & GPS; look into SHP file (also KML files)

**Step 2: Create heat maps for species density - done**
Estimate the sensor range for each modality and generate heat maps showing where animals occur by species.

CSV → Shapefile format conversion (script for this)

**Step 3: Create arrival rates from each modality**

# Meeting Notes

Yuanyang's update to Tanya 3/9/26 (copied from email chain)

- I have downloaded and started working with the camera trap portion of the SmartWilds collection. So far, I have built a simple demo pipeline for organizing the image data, extracting timestamps, constructing event-level tables, and generating some initial summary tables and plots. At this stage, my work is mainly focused on indexing images, creating event records, and carrying out preliminary temporal analysis.
- The detection and modeling part is still at a placeholder/demo stage. I have not yet finalized the model choice or built a complete recognition pipeline for identifying animals or animal groups in each frame. For now, my main goal has been to establish a clean and reproducible analysis workflow and generate some initial outputs.
- My current code repository is here: https://github.com/TIARE006/imageomicsthewilds_cameratraps
- The SmartWilds project collection is here: https://huggingface.co/collections/imageomics/smartwilds
- For the next stage, I would like to look into how other researchers handle multi-sensor wildlife data analysis in similar work, and review related papers to see what ideas or methods may be useful for this project.

Jenna/Yuanyang 3/13/26

- Next steps:
    - Add current repo to imageomics repo - https://github.com/Imageomics/wilds_multimodal_data
    - Analyze full dataset in OSC - make sure Yuanyang has access
        - Data location: /fs/ess/PAS2136/TheWilds2025/thewilds_20250804_20250828
        - Want arrival rates for camera traps, acoustics, drone, citizen science
        - Jenna needs to do some data analysis here
    - Scope out requirements to get data ready for multimodal AI - Devis Tuia's lab has some current papers - start with the MaskSDM one
        - https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/2041-210x.70200
        - https://github.com/zbirobin/MaskSDM-MEE
        - https://zbirobin.github.io/publications/miam/
        - What do we need to do to make our data multi-modal ML ready?
- Off for spring break? Yes - will need after break
- Jenna to set up weekly TB with team